

Memory Management

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```

cudaError_t cudaHostAlloc ( void ** ptr,
                             size_t size,
                             unsigned int flags
                             )

```

Allocates `count` bytes of host memory that is page-locked and accessible to the device. The driver tracks the virtual memory ranges allocated with this function and automatically accelerates calls to functions such as [cudaMemcpy\(\)](#). Since the memory can be accessed directly by the device, it can be read or written with much higher bandwidth than pageable memory obtained with functions such as `malloc()`. Allocating excessive amounts of pinned memory may degrade system performance, since it reduces the amount of memory available to the system for paging. As a result, this function is best used sparingly to allocate staging areas for data exchange between host and device.

The `flags` parameter enables different options to be specified that affect the allocation, as follows.

- **cudaHostAllocDefault**: This flag's value is defined to be 0 and causes [cudaHostAlloc\(\)](#) to emulate [cudaMallocHost\(\)](#).
- **cudaHostAllocPortable**: The memory returned by this call will be considered as pinned memory by all CUDA contexts, not just the one that performed the allocation.
- **cudaHostAllocMapped**: Maps the allocation into the CUDA address space. The device pointer to the memory may be obtained by calling [cudaHostGetDevicePointer\(\)](#).
- **cudaHostAllocWriteCombined**: Allocates the memory as write-combined (WC). WC memory can be transferred across the PCI Express bus more quickly on some system configurations, but cannot be read efficiently by most CPUs. WC memory is a good option for buffers that will be written by the CPU and read by the device via mapped pinned memory or host->device transfers.

All of these flags are orthogonal to one another: a developer may allocate memory that is portable, mapped and/or write-combined with no restrictions.

[cudaSetDeviceFlags\(\)](#) must have been called with the [cudaDeviceMapHost](#) flag in order for the [cudaHostAllocMapped](#) flag to have any effect.

The [cudaHostAllocMapped](#) flag may be specified on CUDA contexts for devices that do not support mapped pinned memory. The failure is deferred to [cudaHostGetDevicePointer\(\)](#) because the memory may be mapped into other CUDA contexts via the [cudaHostAllocPortable](#) flag.

Memory allocated by this function must be freed with [cudaFreeHost\(\)](#).

Parameters:

ptr - Device pointer to allocated memory
size - Requested allocation size in bytes
flags - Requested properties of allocated memory

Returns:

[cudaSuccess](#), [cudaErrorMemoryAllocation](#)

Note:

Note that this function may also return error codes from previous, asynchronous launches.

See also:

[cudaSetDeviceFlags](#), [cudaMallocHost](#), [cudaFreeHost](#)